To understand PLCs, we need to address, in this course, PLC history, architecture, operation, ladder logic programming, applications, and, in particular, electrical controls. PLCs are devices intended to do the same functionality as a relay. So, in another video, in another part of the course, we'll talk about relays. That's relatively quick because you must understand the relay to understand PLCs. Relays are handy control devices, but they do have their problems. In particular, they are electromechanical. Because they are mechanical, they can and will wear out. The PLC overcomes this by virtualizing the relay, so you get the same functionality as a relay but using the PLC without the moving parts, which is a considerable improvement in control technology.

The term "control technology" contrasts with "instrumentation technology." Both deal with automation and control, but the PLC is more concerned with electrical controls in electrical automation, whereas instrumentation is more concerned with analog control in continuous process control. Again, the PLC is in the electrical domain, and on/off type control is what that term implies. There are two types of industry: batch and continuous control. Batch control is the domain of PLCs. Batch is like making a batch of cookies, but continuous control deals with analog. Meanwhile, control, or PLC control, deals with digital signals (on/off status, black and white), while analog signals are black and white and have shades of gray in between signals.

In instrumentation, we talk about 4 to 20 milliamps, with infinite positions or values between 4 and 20 milliamps. The nature of analog is infinity between the lower and upper range value. For example, if you're trying to position a control valve to throttle it, we choose analog because we can go with any position between open and closed. That's analog. That is the realm of instrumentation. Contrast that with batch control and the term batch.

When you think of batch, think about a batch of cookies. If you're making a batch of cookies, you add ingredients individually. You open the flour hopper, dose out so much flour, and close the hopper. You then open the milk hopper and close the hopper after you have the right amount of milk. So you see, the valves are going open and then closed. When you need sugar, you open up the valve that dispenses sugar into the mixing tank, and then you close it when enough sugar has been dispensed. So this is the nature of batch. When you finish adding the ingredients to batch controls, you will mix them. That would involve turning on a mixer or blender and then turning off that device. So, all these actions in the electrical realm of control are on/off devices. Not only that, but they also get information from the surrounding world that is on/off. In this regard, there is a report in this course on input devices, which are switches. There are limit, proximity, hand, photocell, pressure, level, flow, and temperature switches. A switch is a device that tells you information, but it only tells you information at one particular point. It doesn't tell you anything more than the temperature is at 180 degrees because that's the set point for the actuation of the switch. It tells you nothing more. It'll tell you when it crosses 180. It can't tell you how much more heat is in the process other than 180 degrees. It can't tell you how much less. The only thing you can know is that it's either at 180 degrees (possibly higher) or not at 180 degrees and is somewhere lower, but you don't know that value. So, on/off information is the nature of what we feed a PLC to measure the environment around it.

And then again, contrast that with instrumentation, where there's a window to see the temperature. You may have 180 degrees Fahrenheit as the set point, but you can see as low as let's say, 150 degrees and as high as 210 degrees. And then, with that window of view, you can know if the temperature is a little below 180 or a lot below 180. If it's a little higher than 180 or a lot higher, and then it, you know, is it on the set point 180? So, there are two types of control: automation instrumentation and controls. Controls are electrical; instrumentation is instrumentation. Controls are batch, on/off processes; instrumentation is analog process measurement and throttling. So that's important to understand.

And when you talk about these controls, historically, batch controls have been done with the relay. And the relay, again, is something we'll go into in another lecture, but the relay is the precursor to the PLC. The PLC intends to do the functionality of the relay on/off control, but it's electromechanical. So, with the advent of the PLC, you got rid of many maintenance problems, and you essentially have a computer doing the functionality of relays, not just one but many.

PLCs were invented by a company called Modicon for the automotive industry to assemble automobiles. This technology then spread throughout all sectors of the industry. You'll find PLCs in anything from race cars to amusement park ride control, elevator operation, making beverages, and making batches of cookies. Budweiser is a local employer that hires electricians who need to understand how PLCs work because that would be a batch process. That is not to say that PLCs are not used in continuous processes like refineries and chemical plants. They are used more incidentally. In refineries and chemical plants, you hear the term DCS (Distributed Control Systems) more often, and that's another topic for another day, another course. But it's in the realm of analog control, even though it uses digital instrumentation at this present time in many plants.

We're going to go into the architecture of a PLC, and then we'll go into the ladder logic programming and applications. So, I'm going to switch over now to the automationdirect.com website. Automationdirect.com manufactures the Do-more PLC, the Click PLC, and others. It is also the one that gives you a fantastic PLC simulator called Do-more that is suitable not only for programming without a PLC and being able to see the functionality of your program but also that same program allows you to hook up with an information cable, a data highway cord, to transfer what you program into the hardware Do-more PLC.

So now let's go over to sharing the application. I'm going to choose the ELPT 2319 web page. I'm going to click now, and I'll fade away. Now you see infinity in front of you. The web page is basically like being in a room full of mirrors. I will switch over to the Automation Direct web page and let you see. Let's start by going to a different page. We're going to click on the logo, Automation Direct, and help you understand how to navigate this. Now, it's an essential tool to know the manufacturer's website and use it effectively. Here, we see the front page. Whenever you click a logo, it typically takes you to the main page. You can see the AutomationDirect.com home page up here. And also, let's look down to the left. There is a menu for shopping. We see new and trending information here about the products they are promoting. Let's scroll down, and we're going to "Programmable Control." Remember, in another video, I said the old name for PLCs is PCs, and here you see it echoed under "Programmable Control." Let's go ahead and click.

The page is hopping around, but let's look and see here: interactive PLC selector, Arduino compatible controller, so robotics use Arduino and Raspberry Pi, Click PLC, Productivity PLCs, Do-more, Direct Logic (that's a legacy one), motion controllers, I/O links (input-output links), for example. Let's click on "Programmable Controller" and look at that page. Here, we're going to see their line of PLCs. I just reviewed those with you, but the one we're interested in for this course is the Do-more series. Let's get more specific by going to the Do-more.

Here we are on the page for Do-more. In another part of this course, you will learn the methodology for downloading software. I again have a lagging computer here; it's still thinking. As it finishes thinking, we can look and see that there are a whole lot of products listed under here. We're not going to bother with that right now. We're going to choose which of these Do-mores we're going to deal with. Now, there's a new architecture called the Bricks. It uses the Do-more computer processor, which is the more legacy model. The card itself looks gray. It's plugged into a legacy framework, which we're about to go over. The typical precursor to all PLCs is this configuration. Then there's another one, a stackable PLC, which is also new. I will go to this one and give it time to load the page.

Okay, and now here's an excellent tutorial in itself. If I'm dealing with this product line, the Do-more H2 series, it talks about the CPUs, bases, and some cards. PLC architecture is straightforward, as you're going to find out. You need a power supply, a CPU (a computer), and something to plug things into (a base, also known as a rack). In the rack, plug-in I/O (input/output) cards. Those are cards that plug into one of these slots in the base. Here's an AC (alternating current) input-output card that will allow you to bring in switches and send out information that uses DC voltage. Same thing but for AC. Here's one that combines AC and DC input and output. The actuality is that it's saying AC or DC inputs, but the outputs are relay outputs. Now, that's curious because I told you just now that PLCs were created to do the functionality of a relay. So here we learn that PLCs do not do away with relays. Relays are one of the most versatile control devices, and there's a perfectly good reason for using a relay for an output: you can use external power with this PLC without having the PLC supply your output power. Analog input and output show you that PLCs have come a long way from being only for batch control. But PLCs today can do a DCS continuous control configuration functionality by adding appropriate cards. Not only that, but we can deal with Ethernet input and outputs. Some instrumentation on the factory floor uses Ethernet devices and communication cards, which are specialty modules. Programming software is what you're downloading for free. Automation Direct has this zip link wiring solution because they made a very organized and valuable method of tying wires to their PLCs. You see, PLCs are going to take a lot of wires coming in and a lot of wires going out to devices. Spare parts, such as starter kits and the documentation manual, are also listed here, which I'll be entering in just a moment.

When we go down, we can again see more particular information about all these devices. It's clickable for specs, drawings, and more information. Let's click on "software" to help you with the programming software. It will be discussed in another video in another section of the course. I just clicked on "software," this is how you acquire the free, amazingly powerful PLC simulator called Do-more Designer. Down here, some students hear what I say about it being free, and then they look and say, "Wait, wait, wait, wait. It says eleven dollars and twelve dollars here." But that is if you want hard media. But if you don't, just look here for the blue letters, "Download it." It says to download for free. You click there, and it will open up another screen that will lead you down a trail to download the software. Click on "Download Software."

Now, as it goes to the next screen, let me tell you, it will be archived when it downloads. It's going to be zipped, which means it's going to be compressed. You have to decompress the download before you install it. Please do not install it before you decompress it. Again, another place to push "Download." Now, at this point, you need to put in your email address. Yeah, they give you an amazing product for no cost, but as any company, they intend to try to communicate with you, so that's why they email. Put in your lead college email or your email. After all, you will become a technician, and this is a tremendous technical company. Then, once you do, you can click "Download" and download the program. Down at the bottom, I see it wants me to accept cookies. It's a trustworthy website. They certainly do not allow viruses on their website so that you can download a cookie. If you are unwilling to, perhaps you can get this without doing that. That's up to you.

Okay, so now let's get over here to the user manual. Let's go back a couple of steps. One of the clickable links was the manual. The manual can no longer be bought but can be downloaded freely. I've already downloaded it to review the user manual web page here. I clicked on it, so there it is. You can download the entire PDF manual. If you want it, it's about an inch thick, actually more than an inch thick, with all kinds of information. We won't use much of it because this is a foundation course. But since it pops up, let me show you that Do-more offers many video tutorials. It's a brilliant company in that it offers free training. Those may not be so useful to you because they talk about analog temperature and PID (proportional integral derivative), which talks about analog and instrumentation. But there are some videos you may want to download and explore that are made by the company.

So here is the manual. Again, you can download it for free, but once you click on that link, you can download it chapter by chapter. I want to show you chapters one, two, and eight, which are a getting started, Do-more overview of the installation and wiring. Here is chapter one. I will go back to the top of chapter one. Look at what chapter one contains. You can see there's a table of contents on the left here, and you can go into this at any point, but I'm just going to scroll through it. Let's talk about it. There's the table of contents, the manual's Purposes, the chapter's purpose, and some regulatory information. Now, let's look at this PLC. Here, we're going to learn about the architecture of the PLC. It's straightforward.

A PLC rack or base unit with this 110/220 volt AC power supply is here. This power supply gives power to the PLC like any power to a device. You've got to plug it into the wall for it to work. It is the same with the PLC. That power supply also furnishes power for inputs and outputs if you so desire to power those internally. Then, look at the other card next to it; the power supply is on the left. This card order has historically been very standard across brands and manufacturers. Anyway, this second card is the computer, Do-more CPU. Now CPU means central processing unit. If you're a computer techie, you know that's another term for expressing a computer, the actual processor. You can see there are ports on this. There are on, off, and program switch modes. There are some indicator lights. Next are two more slots where you can plug in interface cards. This one is the simulator module. It lets you use these switches instead of wiring up pressure switches, level switches, proximity switches, and photocell switches to program. When you're programming, you want to see what happens if the pressure switch reaches its setpoint, what happens if the lights go off, and the photocell switch activates. It's awkward to try to manipulate all those different types of switches: level, flow, temperature. For programming, using these toggle switches is much easier and more practical. Toggle means it has two positions. You toggle it one way, you toggle it the other way, and that simulates a switch, which is nothing more than making and breaking a wire. On top, the indicator lights tell you whether this switch is activated. On this card, it's an output module. You notice automation direct PLCs have blue tabs for input cards, and output cards have a little red tab or triangle. There is a cover over the wiring terminal block that gives it a decent-looking presentation.

Whenever you finish wiring, you put this little cover over it, and it hides the terminal block. Also, at the top, we learn that the outputs also have an LED indicator to tell us if they're switched on and off. That's it. That's the configuration of a PLC. You can interface with Windows XP, Vista, 7, 8, for example. Of course, by now, it can interface with 10 or 11. Do-more designer software, that's what I just showed you how to acquire.

Here is the data transfer cable. It's called a programming cable, which hooks your PLC to the computer where you'll do your programming. Once the program is complete, you will send the program to the PLC through this information highway so the PLC can have it in the CPU. Some auxiliary things: they'll sell you a screwdriver, which is handy, wire strippers, hookup wire, and an AC cord that you could wire to this power supply. Let's scroll on down. I'm not going to go into all the specs of the PLC. I'm just going to give a walkthrough. If you want in-depth information, download it and read it.

Install Do-more Designer software. There's your installation. If you're having any difficulty after you download it, here's your answer. Launch the Do-more designer software. It's telling you how to get it up and going. It now shows you the features. We won't be doing much on this left window, but this is our programming window here. Notice there's an R rail and some rungs. Those are ladder terms. Rails go up and down both sides of the ladder, and rungs connect the rails. That's where you put your hands and feet as you climb, so you might then understand why they call the programming we're going to do "ladder programming." Ladder programming didn't come out of PLC technology. Ladder programming came out of relay electrical relay controls, and it's a simplified way to see circuitry when you're doing controls.

Install the hardware. It's telling me how to get it going. You can see behind the terminal block now on the output card a legend showing where each of the zero, one, two, three, four, etc., terminals are. You notice this terminal block itself unplugs from the PLC card. You can disconnect it, do all your wiring, and then plug it back in. Connect your PLC to the computer here with the USB cable. Apply power to the Do-more. Now, here, let's look at the wiring strip.

Interestingly, we see a 110/220 VAC base terminal strip, but there's also a 12/24 volts DC base terminal strip and a 125-volt DC base terminal strip. That is the power strip we're going to need to furnish power. As I said, plug it into the wall for power. So it can be done with AC or DC, and when it comes down to DC, if you're not familiar with that, don't plug it into the wall if you have this one because you have to have a source for 125 volts DC. If we look more carefully, that voltage can range from 120 to 240 volts DC. That is not standard. It's not what's in your receptacle in the classroom or at your house. You could plug in this one here, but you notice that if you're in another environment besides standard USA 60 hertz, 120 volts/110 volts, you can still use that and apply it here. Here is 12 or 24 volts. Many pressure switches use 12 or 24 volts. This 12/24 VDC is typical for instrumentation device wiring, and if that's the power supply you have, you can buy a PLC with this specification. Now, these are not alternates to the same terminal block. You have to buy the PLC with this option, or this option, or this option.

Establishing communication between the computer and the PLC is talked about here. Verify hardware configuration. The hardware configuration you notice on this rack, this base, has several more cards than the one we examined previously, which only had the ones populated here: power supply, CPU, input, and output cards. This rack also allows for quite a few more. One, two, three, four, five, and six more cards will plug in.

There is no specification on where you plug the inputs and the outputs. You may want to combine four input cards and then put in your output cards. You may want to alternate input-output, but that's an engineering choice. Yet, the PLC needs to understand the final arrangement. Once it's determined engineering-wise, you do not want to change the configuration, the hardware configuration, or the order of the input-output cards. But once engineered and the configuration is known, you need to let the PLC go out and map the placement of all the cards. It does it automatically so it knows where an input is and where an output is. Historically, it's going to be octal. You'll have maybe zero to seven for the first input card and eight to fifteen for the second. The PLC knows that even though the cards are identical, the one on the left will be the first to give it an address of zero through seven. When you plug in the second one, it knows zero through seven has been used, automatically assigning eight through fifteen to the second card. Octal is interesting because it's a power of two, and you'll see that theme in this course: binary, octal, decimal, and hexadecimal. We'll be talking about all these number systems.

Again, getting started, I'll let you read this as you see fit. Create a ladder logic program. I'll walk you through exactly how to do this, but it never hurts to have another way of explaining it. Here, you see they have put a rung into place, a couple of rungs. We have a contact here. Notice there's a space in between; that's a "normally open" contact. You draw it as "normal," so that's open. If it were closed, there'd be a line dissecting it, telling you that electrons can go through it if it's "normally closed." But this one, electrons, is the power source of this left rail, and electrons can get to here, but they can't get to the right side because that's a gap. It's the same thing down here. When this loses, the power gets to this point and energizes this timer, which meets its set point after three seconds, and all the contacts it owns switch. This rundown shows you what we do in this course, giving you a little overview.

Introduction. How to save a project. It is not enough to program using your Windows interface. That program is not in the PLC until you write your project to the PLC. It points out this little button. You can save your program to the Windows interface, and that's an intelligent thing to do, especially if you have a power outage and don't want to lose all your work. But when you finally want to try your PLC programming, you must write it to the PLC. Extensive online help exists in the program.

Now, I'll go to chapter two, a little briefer. It's a PLC overview. Let's look at the architecture again. Here, I am showing you the Do-more H2 series on the computer. You can see a connector resembling a Cat5 and other connectors like a telephone jack. That plugs into this slot only; you would never put the PLC computer anywhere but in this second slot. The first slot requires the power supply. It shows system compatibility, base units, discrete input-output, analog, and specialty modules. Look at this PLC's versatility—another chart showing communication protocols. You see Modbus. Remember I mentioned that Modicon is a manufacturer of PLCs that invented PLCs? They're still around, but there's Direct Logic, Direct Net, and Direct Logic's protocol. You see the Do-more. That's it for this chapter.

The last chapter I briefly want to look at is chapter eight, which is about wiring. We're going to leave wiring to PLC, too. It's enough for us to understand what a PLC is, gain the foundational knowledge of a PLC, and learn how to program a PLC in this course. A PLC mounts in a box, so it's suitable for being put into an enclosure. Observe specific considerations when you panel mount it or put it into an enclosure. It's going to tell you about those. You can see a ground bus here for grounding electricity. You can see a plate it's mounted on, the plate mounted in the enclosure. That is a very standard installation. It can be mounted outside. It could be mounted inside the control center as well. However, PLCs are usually rugged enough to withstand the conditions inside a protective case and enclosure.

When installing the bases, you learn that this PLC can snap onto a DIN rail. You hook the top part on the back channel, swing it, and then push it; it locks in. This rail-mounted terminal block end cap shows you how to swing it and lock it into place. Very easy to install. You install this rail first, then snap on the PLC, and you can snap on terminal blocks as well—the same DIN, same technology. We've seen that before.

Now, here are some wiring basics. We've seen that before in the other module and chapter. I/O modules, position, and wiring. Here, look at your input-output modules: DC input, AC input, DC, AC output, relay output, analog input instrumentation, etc. Specialty modules. PLCs are excellent devices; this is the best Christmas present ever for techies. The only thing you want Santa to bring you is the PLC, which, by the way, you can acquire for about $120 with about, I think, six outputs and eight inputs, something like that; that's Very cheap. But it would help if you had devices. It would help if you had switches for inputs and things to control. The things to control are motors and pumps, solenoid valves, dampers, positioners, and lights. I/O checklist. Module type. On this one, you get four points for your input, or eight, twelve, or sixteen. How big is the wire diameter for wiring, and how much force is required to torque it down? Very techy stuff here. It shows you here that all your inputs and outputs should be fused. Here's an example of snap-on DIN rail fuses. Ziplink is their wiring protocol. You can read about that if you like. More about wiring isolation boundaries. Wiring is not the nature of this course, so let's pass it. Scott Churchman sent me the link to a couple of videos I placed in the course. It's in the optional information about sinking and sourcing. So, if you want a head start on wiring PLCs, it would be good to watch those videos and get that thought about sinking and sourcing. It goes into a tutorial here about that.

Input-output common terminal concept. You have common wires and unique wires. Common is going to be common to every switch. Observe how it goes from this common. Here's your external power supply. The common is going here, which is to the negative side of the battery. But on the positive side, you have these unique wires. They're taking off in parallel, and one wire goes to every different kind of switch: decimal switch, symbol, pressure switch, level switch, hand switch, whatever switch. So, in wiring a PLC, the term common wire is used. In this case, it's negative, and the positive wire is unique. Whenever this switch closes, it goes to a unique input terminal of the PLC, so the PLC knows whether that switch is "made." These are symbols for wiring: AC, DC, AC or DC power supply, and input switches. They're telling you their symbology: solid-state input sensors, output loads.

Outstanding manufacturers tend to have excellent tutorials on using their equipment, and Automation Direct is no different. If anything, they go overboard and give you sound technical information for free. And that's it. I'm not going into anything else in this video. It's been long enough. I hope you have had a chance to understand what's said here and got some information out of it that will be helpful to you. We'll see you in the next video.